

MPG No. 23.3.84 $6'/m$ [Type III, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxz} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xyz} & -M_{xyz} & 0 & 0 & 0 & -M_{xxz} \\ M_{xxz} & -M_{xxz} & 0 & 0 & 0 & M_{xyz} \\ 0 & 0 & 0 & -M_{xxz} & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = M_{gb}^{(1)}$$

$$M_{xxz} = M_{ga}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxz} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xyz} & M_{xyz} & 0 & 0 & 0 & M_{xxz} \\ -M_{xxz} & M_{xxz} & 0 & 0 & 0 & -M_{xyz} \\ 0 & 0 & 0 & -M_{xxz} & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = 2T_{f2}^{(1)}$$

$$M_{xxz} = -2T_{f1}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xyz} & M_{xxz} & 0 \\ -M_{xyz} & -M_{xxz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxz} & -M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = T_{f2}^{(2)}$$

$$M_{xxz} = T_{f1}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzx} & -M_{yzx} & 0 & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = T_{f2}^{(3)}$$

$$M_{yzxy} = T_{f1}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxz} & 0 & M_{xxzy} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxz} & 0 & -M_{xxzy} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzx} & -M_{yzx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & -M_{xxz} & M_{xxzy} & 0 & -M_{xxzx} & M_{xxzy} & 0 \end{bmatrix}$$

$$M_{xxyz} = M_{gb}^{(1)} + 2T_{f2}^{(1)} + T_{f2}^{(2)}$$

$$M_{xxzx} = M_{ga}^{(1)} - 2T_{f1}^{(1)} + T_{f1}^{(2)}$$

$$M_{xxzy} = M_{gb}^{(1)} + 2T_{f2}^{(1)} - T_{f2}^{(2)}$$

$$M_{xxzx} = M_{ga}^{(1)} - 2T_{f1}^{(1)} - T_{f1}^{(2)}$$

$$M_{yzxx} = M_{gb}^{(1)} - 2T_{f2}^{(1)}$$

$$M_{yzxy} = -M_{ga}^{(1)} - 2T_{f1}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = T_{f2}^{(3)}$$

$$M_{yzxy} = T_{f1}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxzx} \\ 0 & 0 & M_{yyyy} \\ 0 & 0 & 0 \\ -M_{xxzx} & M_{yyyy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyy} \\ 0 & 0 & 0 \\ M_{xxzx} & -M_{yyyy} & 0 \\ 0 & 0 & -M_{xxzx} \\ -M_{yyyy} & -M_{xxzx} & 0 \end{bmatrix}$$

$$M_{xxzx} = M_{ga}^{(1)}$$

$$M_{yyyy} = -M_{gb}^{(1)}$$